6.4.2 Extension of range

a) DC voltmeters

A DC milli ammeter may be converted as a voltmeter by connecting a high resistance called multiplier in series with the instrument. The multiplier limits the current through the meter so that the current through it not exceed the value of full scale deflection current. A voltmeter measures the voltage across two points and hence connected in parallel to the terminals.

Let,

R_s = multiplier resistance

I_m = full scale deflection current of the instrument

R_m = resistance of the moving coil of the instrument

V = voltage of the circuit to be measured

$$V = I_m R_m + I_m R_s$$

$$I_m R_s = V - I_m R_m$$

$$\therefore R_s = \frac{V - I_m R_m}{I_m}$$

Do you Know?

Ammeters must always be connected in series in a circuit and have a very low internal resistance. Voltmeters must always be connected in parallel in a circuit and have a very high internal resistance

b) DC ammeters

The moving coil of PMMC instrument is light and small and hence can carry very small current. The galvanometer may be converted as an ammeter by connecting a very low value of resistance called shunt in parallel with the instrument. The shunt is used to bypass major part of current so that the current through the moving coil not exceed the value of full scale deflection current. An ammeter measures the current through the circuit and hence connected in series to the load.

Let,

R_{sh} = the value of shunt resistance

I_{sh} = current through the shunt

I_m = full scale deflection current of the instrument

R_m = resistance of the moving coil of the instrument

I = current of the circuit to be measured

$$I_m R_m = I_{sh} R_{sh}$$

$$\therefore R_{sh} = \frac{I_m R_m}{I_{sh}}$$

$$R_{sh} = \frac{I_m R_m}{I - I_m}$$

6.4.3 Moving iron type instruments

The moving iron instruments are used to measure both AC and DC voltages and currents. They are classified into two types:

(i) Attraction type
(ii) Repulsion type

(i) Attraction type moving iron instruments

The construction of the attraction type instrument is shown in the figure 6.7. Attraction type moving iron instrument

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